AMERICAN HERITAGE SCHOOLS
COMPETITIVE MATHEMATICS

2024 Mock Theta Individual #2

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1. Compute $||||||1-5|+3|-2|+10|-7|+15|-24|+60$.

2. It takes 3 Thetas exactly 3 hours to fail 6 tests. How many hours does it take for 1 Theta to fail 1 test?

3. Denote the number of possible solutions given by the Rational Root Theorem for the following equation as N:

$$f(x) = 8x^6 - 10x^4 + 9x^7 + 2x^2 - 3x^2 + 2x - 12$$

Find the number of (positive, integer) divisors of N

(A) 0 (B) 36 (C) 59 (D) 60 (E) NOTA

4. Ishaan has decided to drop Calculus and live the life of a farmer, just as he's always wanted; however, he can't help but reminisce about his favorite part of Calculus: optimization. Ishaan has 20 feet of fencing and wants to build a pen for his cows. What is the largest possible area that he can enclose using only his 20 feet of fencing?

5. Meng runs at a speed of x miles per hour to visit her family in China. When returning to the States to get a perfect score on all the Geometry topic tests, she runs at a rate of $3x + 20$ miles per hour. What is, in terms of x, Meng's average speed for the round trip?

6. Liz and Ellie live on a isolated 2×3 grid, with Liz's house at the bottom left and Ellie's at the top right. This grid floats gently on lava, with only the grid's lines visible (basically, no shortcuts allowed!). Assuming she only travels up or to the right (and that living is a priority to her), how many ways can Liz get to Ellie's house?

7. Bennett has an odd function $f(x)$ and Krishang has an even function $g(x)$. Together, they use their functions to make a new, better function, but they want it to be even. Which of the following is even?

(A) $f(x)g(x)$ (B) $2f(x)$ (C) $f(x)(g(x))^2$ (D) $(f \circ g)(x)$ (E) NOTA

8. How many asymptotes does the following function have?

$$f(x) = \frac{-e^{2x} + 7e^x - 10}{e^{2x} - 5e^x + 6}$$

9. Gui's *Yu-Gi-Oh!* deck consists of 40 cards, 5 of which are the *pieces of Exodia*. If Gui has these 5 cards in my hand at any given time, then he win the duel. In *Yu-Gi-Oh!*, a player's opening hand consists of 5 cards, each drawn from the top of the deck. What is the probability that Gui draws all 5 *pieces of Exodia* for his opening hand and instantly win the duel? Assume that the deck is randomly shuffled.

(A) $\frac{5}{40!}$ (B) $\frac{5!}{40!}$ (C) $\frac{35!}{40!}$ (D) $\frac{5! \times 35!}{40!}$ (E) NOTA

10. Kyra forgot everything about matrices! Thankfully, she is told by a wise man who goes by "Gooey" that for any matrix M and where $\text{adj}(M)$ is the adjoint matrix of M,

$$M^{-1} = \frac{\text{adj}(M)}{\det(M)}$$

If M is 4×4 and $\det(M) = 7$, find $\det(\text{adj}(M))$.

(A) 1 (B) 7 (C) 49 (D) 343 (E) NOTA

11. What's the remainder when the function $5x^4 - 4x^2 + 3x - 9$ is divided by the function $x^2 - 3x + 4$?

12. Tiger is organizing an epic sponsor *Magic: The Gathering* tournament! He has bought 5 distinct decks of cards, each for one of Mr. Ariew, Ms. Alfonso, Mr. Rovere, Dr. Santos, and Mr. Ysimura. When Tiger shuffles each person's assigned deck, he gets a little bit careless and doesn't check whether or not he gives the set of shuffled cards back to the right person. What is the probability he ends up giving no sponsor their original deck?

13. In Planet Darsh, fair dice have 4 sides labeled 1, 2, 3, and 4. Jiggy the Alien grabs a random, integer number of fair dice and throws them, getting a sum of 4. What is the probability that Jiggy grabbed 2 or 3 dice? Assume that Jiggy is equally likely to grab any number of dice.

14. Suppose that x and y are real numbers that satisfy the following equations:

$$3^x + \sqrt{3}^x = 20$$

$$8^y + 2^y = 350$$

Find $\log_2(3^x) + 4^y$.

(A) 53 (B) 51 (C) 23 (D) 11 (E) NOTA

15. If $x^2 - \frac{1}{x^2} = 2$, find $x^{12} + 3x^4 + \frac{3}{x^4} + \frac{1}{x^{12}}$.

(A) 8 (B) 64 (C) 216 (D) 512 (E) NOTA

16. Compute

$$\sum_{i=1}^{\infty}\sum_{j=1}^{\infty}\frac{1}{3^i}\cdot\frac{1}{5^j}$$

17. $\triangle GUI$ has incenter S and $GU = 15$ and $GI = 13$. Points B and L are on GU and GI respectively such that B, L, and S are co-linear and BL is parallel to UI. Find the perimeter of $\triangle GBL$.

18. Jonathan the Prince must meet Julia the Princess at the top of the cylindrical tower they live in. This tower is 30 feet tall and has a radius $\frac{10}{\pi}$ feet. Jonathan is at the base of this tower and Julia is directly 30 feet above him, so to get to her, Jonathan walks on the surface of this cylindrical tower, not changing the direction he's facing after he starts moving, and revolves around it twice before getting to Julia. How many feet did Jonathan walk before reaching Julia?

19. Let $RAJM$ be an isosceles trapezoid with a perimeter of 104 with bases RA and JM and $RB > RM$. There exists a point M' on RA such that $RM = RM'$. The altitude from R to MM' intersects JM at R'. IF $AJ = 17$, find the perimeter of $M'AJR'$.

20. Let $F_0 = F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for all integers $n > 1$. Find

$$\sum_{n=1}^{\infty}\frac{F_n}{5^{n+1}}$$